PCA

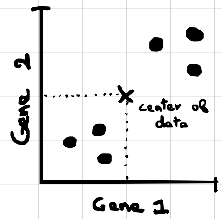
PCA for dimensionality reduction

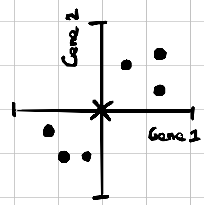
1-3 variables are easy to plot but 4+ requires impossible higher-dimensional space.
PCA simplifies high-dimensional data into a 2D or 3D plot to spot clusters or similar points.

How PCA works? (with SVD)

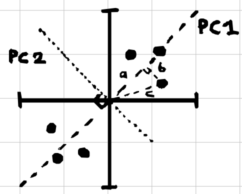
1. centering the data to origin:
   - mean of variable j : $\mu_j = \frac{1}{n} \sum x_{ij}$

     centered : $X_{ij} = X_{ij} - \mu_j$



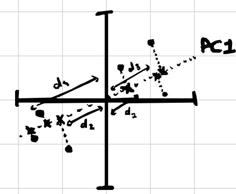


2. PCA finds the best fit line :
   - maximize the distance c from the origin (same as min of b) $(a^2 = b^2 + c^2)$
   - slope of this line is PC1, perpendicular to it is PC2.

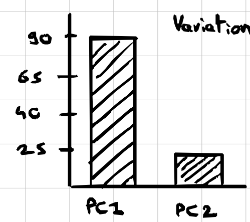


PC1 has slope of 1.
PC1 eigenvector is $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$
$\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ are the loading scores

3. Compute Eigenvalue and Singular value :
   - Eigenvalue PC1 : $\frac{1}{n-1} \sum (\text{distance to PC1})^2$
   - Singular value for PC1 : $\sqrt{\sum (\text{dist. to PC1})^2}$



4. Total variation of PC1 and PC2 :
   - $\frac{1}{n-1} \sum (\text{dist. to PC1})^2$ = variance PC2
   - $\frac{1}{n-1} \sum (\text{dist. to PC2})^2$ = variance PC2



Scree plot shows how much variation is explained by each PC.

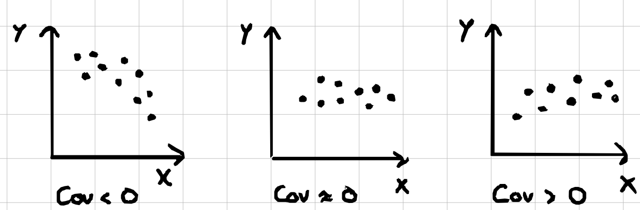
Using covariance matrix

First step is the same we center the data to (0,0)

- Compute the covariance matrix:

$$\Sigma = \begin{bmatrix} \text{Variance}(X) & \text{Cov}(X,Y) \\ \text{Cov}(X,Y) & \text{Variance}(Y) \end{bmatrix}$$

sum of product of coordinates



First, this matrix is always symmetric.
Second, this linear transformation is always a stretching accross two eigenvectors.

- Solving for Eigenvalues, we get to know which axis has higher variance. (PC1 = eigenvector with max eigenvalue)
- Choose 2-3 eigenvectors with highest eigenvalues to represent data in 2D, 3D.